

WEEK 7 - TESTING & VALIDATION EVIDENCE

Data Analytics Track | Evidence Register and Validation Record

PURPOSE

This document provides the evidence register for Week 7 testing and validation. It links each test to the Week 6 baseline, records the observed/documented result, identifies the evidence available, and clearly separates completed checks from tests that could not be completed with the supplied evidence.

1. Evidence Baseline

The Week 6 Advanced Analytics output is the baseline for Week 7 testing. The supplied Week 6 report and dashboard establish the following documented baseline:

Baseline item	Documented value / finding	Evidence status
Analytical population	5,000 appointments	Documented in Week 6 report/dashboard
No-shows	2,423	Documented in Week 6 report/dashboard
Overall no-show rate	48.46%	Documented and mathematically reproducible from the stated numerator/denominator
Previous no-shows	Observed increase across the 0–3 grouping	Documented Week 6 visual/report
Booking lead time	24.18% at 0–2 days to 60.49% at 31+ days	Documented Week 6 visual/report
Reminder channel	None 51.39%; WhatsApp 49.77%; Email 48.41%; SMS 45.75%	Documented Week 6 visual/report
Appointment type	Four rates relatively close	Documented Week 6 interpretation
Key takeaway	Longer booking lead times and previous no-shows are the strongest observed indicators of non-attendance.	Documented Week 6 dashboard

2. KPI Validation Evidence

Test: reconcile the headline no-show KPI using the documented Week 6 numerator and denominator.

Test step	Evidence / calculation	Result
Population	5,000 appointments	PASS - agrees with supplied Week 6 baseline.
No-show numerator	2,423 no-shows	PASS - agrees with supplied Week 6 baseline.
Rate calculation	$2,423 \div 5,000 \times 100 = 48.46\%$	PASS - calculation reproduces the documented KPI.
Dashboard KPI	48.46% No-Show Rate	PASS - agrees with the documented calculation.

VALIDATED RESULT

The 48.46% headline KPI is supported by the supplied Week 6 evidence and the stated numerator/denominator calculation.

3. Analytical Validation Evidence

3.1 Previous No-Shows

Week 6 deliberately focused the visual on the interpretable 0–3 previous-no-show groups because the higher categories contain very small numbers of patients.

Previous no-shows	Documented no-show rate	Validation observation
0	43.51%	Baseline group; lowest of the main 0–3 groups.
1	~53.49%	Higher than 0.
2	~59.36%	Higher than 1.
3	67.95%	Highest of the main 0–3 groups.

TEST RESULT

The documented directional pattern is internally consistent: observed no-show rates increase as previous no-show history increases across the main 0–3 groups. This supports retaining previous no-shows as a candidate risk-stratification variable, not as proof of causation.

3.2 Booking Lead Time

Booking lead-time band	Documented no-show rate	Validation observation
0–2 days	24.18%	Lowest documented rate.
3–7 days	30.05%	Higher than 0–2 days.
8–14 days	33.55%	Higher than 3–7 days.
15–30 days	43.21%	Higher than 8–14 days.
21+ / 31+ days	60.49%	Highest documented rate; use the exact band label shown in the final dashboard/report.

TEST RESULT

The documented lead-time rates show a clear increasing pattern from the shortest to the longest booking window. The Week 6 report calculates a 36.31 percentage-point difference between the shortest and longest bands.

3.3 Reminder Channel

Reminder channel	Documented no-show rate	Validation interpretation
None	51.39%	Highest observed rate.

WhatsApp	49.77%	Intermediate observed rate.
Email	48.41%	Intermediate observed rate.
SMS	45.75%	Lowest observed rate.

REFINEMENT / CONTROL

The lower observed SMS rate is not treated as evidence that SMS causes better attendance. Reminder assignment may be related to other characteristics or operational factors. The appropriate Week 7 conclusion is that reminder channel is a candidate for further controlled/statistical testing.

3.4 Appointment Type

The Week 6 report states that the appointment-type no-show rates are relatively close. The Week 7 validation therefore retains appointment type as contextual evidence rather than presenting it as a strong standalone risk indicator.

4. Dashboard Testing Evidence

Dashboard component	Validation check	Status	Evidence / action
KPI card	Does the displayed KPI agree with the documented calculation?	PASS	48.46% agrees with 2,423 / 5,000.
Appointment Type chart	Does the visual support the documented interpretation?	REFINED	Added 48.46% overall benchmark line to improve comparison with the overall rate.
Previous No-Shows chart	Does the visual show the documented 0–3 pattern?	REFINED	Improved label positioning and added a trend line to make the progressive pattern visible.
Booking Lead Time chart	Does the visual show the documented gradient?	REFINED	Added 48.46% overall benchmark line to highlight the 31+ day rate of 60.49%.
Reminder Channel chart	Does the wording avoid causal interpretation?	REFINED	Renamed to Observed No-Show Rates and refined interpretation to avoid causal claims.
Key Takeaway	Does it reflect the strongest documented indicators?	PASS	Retained and aligned with validated observed indicators.
Business Insight/Action panels	Are actions connected to findings?	PASS	Actions address scheduling, prior no-shows and reminder testing.

5. Before → After Refinement Record

Area	Week 6 baseline	Week 7 refinement
Dashboard baseline	Week 6 dashboard showed the four analytical visuals and business interpretation.	Duplicated as a separate Week 7 page so the Week 6 baseline remained unchanged.
Appointment Type	Four rates shown without an explicit overall benchmark.	Added 48.46% overall no-show benchmark line to improve segment comparison.
Booking Lead Time	Five lead-time bands showed the increasing pattern.	Added 48.46% overall benchmark to make the 31+ day 60.49% rate stand out against the overall rate.
Previous No-Shows	Bars showed increasing no-show rates across 0–3 previous no-shows.	Improved data-label positioning and added an upward trend line to make the progression clearer.
Reminder Channel	Reminder channel rates were shown and SMS had the lowest observed rate.	Renamed visual to “Reminder Channel: Observed No-Show Rates” and revised interpretation to distinguish association from causation.
Week 7 validation layer	No explicit Week 7 validation header/panel.	Added Week 7 Testing & Refinement header and validation panel summarising the validated KPI and principal findings.

6. Test → Finding → Action → Retest Record

Test	Finding	Action / refinement	Retest / validation status

KPI reconciliation	48.46% reproduced from 2,423 / 5,000.	Retain KPI.	Validated against supplied Week 6 evidence.
Appointment type visual test	Rates remain relatively close; differences are contextual rather than a strong standalone targeting basis.	Add 48.46% benchmark line.	Retested visually; benchmark displays correctly and improves comparison.
Previous no-show visual test	Rates increase across 0–3 previous no-shows.	Improve labels and add trend line.	Retested visually; upward progression is clearer without changing underlying values.
Lead-time visual test	Rates increase from 24.18% at 0–2 days to 60.49% at 31+ days.	Add 48.46% benchmark line.	Retested visually; longest lead-time group is clearly above the overall rate.
Reminder channel interpretation test	SMS has the lowest observed rate, but causality is not established.	Rename to observed rates and revise insight/action wording.	Retested visually and textually; causal interpretation removed.
Dashboard page structure	Week 7 must show meaningful forward progress from Week 6.	Duplicate Week 6 page and create a dedicated Week 7 Testing & Refinement page.	Validated: Week 6 remains as baseline and Week 7 contains the refinements.
Cross-track test	No collaborator/model output received.	Do not claim completed cross-track testing.	NOT COMPLETED - requires actual collaborator test.

7. Cross-Track Evidence Status

IMPORTANT

The Week 7 requirement calls for a meaningful cross-track testing activity. No completed collaborator output was received, so no joint Test → Finding → Action → Retest → Validated Improvement is claimed in this evidence register.

Evidence item	Available?	How it is used
Collaboration outreach / communication	Yes - previously documented	Evidence that collaboration was attempted; not counted as completed testing.
Data Science model/output	No	No model validation claimed.
Joint dataset/test	No	No cross-track analytical result claimed.

Validated cross-track improvement	No	Marked outstanding.
Data Analytics hand-off	Yes	Candidate variables and testing questions documented for future validation.

8. Outstanding Tests

- Raw dataset reconciliation, if the approved HealthConnect dataset becomes available.
- Live Power BI execution of filters/calculations if required by the submission assessor.
- Stability testing on additional/new data.
- Statistical or controlled evaluation of reminder-channel effects.
- Predictive validation of previous no-shows and booking lead time.
- Meaningful cross-track Test → Finding → Action → Retest activity with another HC-POD track.

9. Evidence to Attach to the Submission

Evidence	Recommended attachment / location	Purpose
Week 6 PBIX	Healthconnect Week 6 Data Analytics.pbix	Primary analytical/dashboard supporting file.
Week 6 dashboard screenshot	Final Week 6 Advanced Analytics dashboard PNG/PDF	Visual proof of KPI, four charts, Key Takeaway and insight/action panels.
Week 7 Testing & Refinement Report	Week 7 report	Detailed testing/refinement narrative.
Week 7 Project Summary	Project Summary document	Convenient end-to-end submission summary.
Cross-track communication evidence	Slack screenshot / saved evidence where appropriate	Shows attempted collaboration; do not label it completed testing.
Week 7 Testing & Validation Evidence	This document	Evidence register linking tests, findings, refinements and outstanding items.
Week 7 final dashboard screenshot	Final Week 7 Testing & Refinement dashboard PNG	Visual evidence of the completed refinements, validation panel and updated interpretations.

10. Final Evidence Statement

SUBMISSION-READY STATEMENT

The Week 7 Data Analytics evidence demonstrates that the documented Week 6 KPI and principal analytical patterns were reviewed and that the dashboard interpretation was refined to distinguish observed associations from causal claims. The evidence also transparently identifies raw-data, predictive and cross-track tests that remain outstanding. No uncompleted collaboration or model validation is represented as completed work.

The Week 7 Data Analytics evidence demonstrates forward progress from the Week 6 dashboard through documented testing and dashboard refinement. The final Week 7 page preserves the Week 6 baseline while adding benchmark

comparisons, a trend-based visual refinement, clearer observed-rate wording and an explicit validation layer. The underlying KPI and documented analytical values were not changed. Raw-data, predictive and meaningful cross-track tests remain transparently identified as outstanding where supporting evidence was not available.